

01.1 Search Strategy

Databases, terms, rounds, and supplementary methods

This document records the search strategy for the A Safe Bet scoping review. The search ran in three rounds between May 2021 and May 2024 across PsycINFO (Ovid), MEDLINE (Ovid), and Web of Science Core Collection. Strings were structured around four conceptual domains (Table 01.1.2) and refined across rounds to improve precision without losing sensitivity. The final strings were validated against an a priori reference set of high-relevance papers.

Note. N = 1,876 unique records screened. Records retrieved in more than one round were credited to the first round. Where a record re-surfaced under refined terms in a later round, the most recent screening decision was treated as final. Strings shown are the Round 3 versions; see Table 1 in the main paper for round-level changes and 01.4 for the PRISMA-ScR flow diagram.

Search rounds

Table 01.1.1 Search Rounds, Periods, Databases, and Yields

Round	Period	Databases	New records	Purpose
1	May–September 2021	PsycINFO (Ovid), MEDLINE (Ovid), Web of Science	1,040	Initial broad search establishing baseline coverage of the literature
2	October 2022–January 2023	PsycINFO (Ovid), MEDLINE (Ovid), Web of Science	218	Refined strings with database-specific terminology and structured exclusion terms; captures 2021–2022 literature
3	February–May 2024	PsycINFO (Ovid), MEDLINE (Ovid), Web of Science	333	Final update; captures 2023–2024 literature prior to taxonomy completion
Unassigned	n/a	n/a	285	Records whose round of entry could not be unambiguously confirmed against publication year (see 01.5 screening record)
Total			1,876	

Note. ‘New records’ counts unique records first identified in that round, after de-duplication against records retrieved in earlier rounds. Records with publication years inconsistent with their round-of-entry label could not be reliably assigned and are reported here as ‘Unassigned’ (see 01.5 for record-level data).

Search domain structure

Table 01.1.2 Conceptual Domains Used to Structure Search Strings

#	Domain	Description
1	Gambling population and risk	Captures gambling activity across the full risk spectrum, combining gambling terms (gambling, bet, wager) with risk and population qualifiers (problem, risk, harm, addiction, recreational, positive play, pathological).
2	Digital delivery context	Restricts retrieval to online, internet, digital, web-based, and electronic gambling settings.
3	Prevention and policy framing	Captures responsible gambling framing through prevention, protection, regulation, and policy terminology (including consumer protection, harm minimisation, risk management, and public health).

#	Domain	Description
4	Intervention tools and techniques	Captures specific tool types, behaviour-change techniques, and therapeutic interventions (for example, deposit limits, self-exclusion, pre-commitment, pop-up messages, feedback, forced breaks, cognitive-behavioural therapy, motivational interviewing).

Database 1: PsycINFO

Ovid APA PsycInfo, 1806 to search date.

Field tags. / = controlled subject heading; ti = title; ab = abstract; id = key concepts (PsycINFO indexer-added terms supplementing subject headings); ADJn = word distance of maximum n words.

Table 01.1.3 PsycINFO Round 3 Search Strings

Line	Search string
1	Gambling/ or Gambling Disorder/ or Pathological Gambling/ or Digital Gambling/ OR ((responsib* or problem* or risk* or harm* or addict* or escalat* or disorder* or recreation* or “positive play” or pathological) adj3 (gambl* or bet or betting or wager*)).ti,ab,id.
2	Internet/ OR online behavior/ OR (online OR internet OR digital OR web* OR electronic*).ti,ab,id.
3	Safety/ or Prevention/ or “Public Health”/ or “Consumer Protection”/ or “Consumer Behavior”/ or “Risk Management”/ or “Protective Factors”/ or “Policy Making”/ or “Harm Reduction”/ OR (responsib* OR regulat* OR protect* OR prevent* OR safe* OR polic* OR framework OR guid* OR recommend* OR strateg* OR “consumer protection” OR “customer protection” OR “risk management” OR “harm minimization”).ti,ab,id.
4	Intervention/ or “Early Intervention”/ or Treatment/ or “Behavior Change”/ or “Digital Interventions”/ or “Health Promotion”/ or Messages/ or Prompting/ or “Warning Labels”/ or “Online Therapy”/ or “Health Behavior”/ OR (tool* OR measure* OR interven* OR task* OR therap* OR prompt* OR “pop-up*” OR treat* OR limit* OR messag* OR nudg* OR cue* OR “behavior change” OR feedback OR counselling OR “self-exclusion” OR “deposit limit*” OR “pre-commitment” OR “forced break*” OR “reality check” OR “proactive contact” OR “brief intervention” OR “motivational interview*” OR “cognitive behav*” OR “structural modification*” OR “low-risk guideline*” OR “self-regulation” OR notification*).ti,ab,id.

Database 2: MEDLINE

Ovid MEDLINE, including epub ahead of print, in-process & other non-indexed citations and Ovid MEDLINE Daily, 1946 to search date.

Field tags. / = MeSH (Medical Subject Heading); ti = title; ab = abstract; kf = author-supplied keywords; ADJn = word distance of maximum n words.

Table 01.1.4 MEDLINE Round 3 Search Strings

Line	Search string
1	Gambling/ OR ((responsib* or problem* or risk* or harm* or addict* or escalat* or disorder* or recreation* or “positive play” or pathological) adj3 (gambl* or bet or betting or wager*)).ti,ab,kf.
2	Internet/ OR (online OR internet OR digital OR web* OR electronic*).ti,ab,kf.
3	Safety/ OR “Safety Management”/ OR “Primary Prevention”/ OR “Secondary Prevention”/ OR “Public Health”/ OR “Health Knowledge, Attitudes, Practice”/ OR “Consumer Product Safety”/ OR “Health Behavior”/ OR “Risk Management”/ OR “Protective Factors”/ OR “Policy Making”/ OR Policy/ OR “Government Regulation”/ OR Guideline/ OR (responsib* OR regulat* OR protect* OR prevent* OR safe* OR polic* OR framework OR guid* OR recommend* OR strateg* OR “consumer protection” OR “customer protection” OR “risk management” OR “harm minimization”).ti,ab,kf.

Line	Search string
4	“Internet-Based Intervention”/ OR Therapeutics/ OR “Behavior Therapy”/ OR “Health Promotion”/ OR “Risk Reduction Behavior”/ OR (tool* OR measure* OR interven* OR task* OR therap* OR prompt* OR “pop-up*” OR treat* OR limit* OR messag* OR nudg* OR cue* OR “behavior change” OR feedback OR counselling OR “self-exclusion” OR “deposit limit*” OR “pre-commitment” OR “forced break*” OR “reality check” OR “proactive contact” OR “brief intervention” OR “motivational interview*” OR “cognitive behav*” OR “structural modification*” OR “low-risk guideline*” OR “self-regulation” OR notification*).ti,ab,kf.

Database 3: Web of Science Core Collection

Citation indexes: Science Citation Index Expanded (SCI-EXPANDED) 1975 onwards; Social Sciences Citation Index (SSCI) 1975 onwards; Arts & Humanities Citation Index (A&HCI) 1975 onwards; Emerging Sources Citation Index (ESCI) 2015 onwards.

Field tag. TS = topic (includes title, abstract, author keywords, and Web of Science Keywords Plus).

Table 01.1.5 *Web of Science Round 3 Search Strings*

Line	Search string
1	TS=((“responsib*” OR “problem*” OR “risk*” OR “harm*” OR “addict*” OR “escalat*” OR “disorder*” OR “recreation*” OR “positive play” OR “pathological”) AND (“gambl*” OR “bet” OR “betting” OR “wager*”))
2	TS=(“online” OR “internet” OR “digital*” OR “web*” OR “electronic*”)
3	TS=(“responsib*” OR “regulat*” OR “protect*” OR “prevent*” OR “safe*” OR “polic*” OR “framework” OR “guid*” OR “recommend*” OR “strateg*” OR “consumer protection” OR “customer protection” OR “risk management” OR “harm minimization”)
4	TS=(“tool*” OR “measure*” OR “interven*” OR “task*” OR “therap*” OR “prompt*” OR “pop-up*” OR “treat*” OR “limit*” OR “messag*” OR “nudg*” OR “cue*” OR “behavior change” OR “feedback” OR “counselling” OR “self-exclusion” OR “deposit limit*” OR “pre-commitment” OR “forced break*” OR “reality check” OR “proactive contact” OR “brief intervention” OR “motivational interview*” OR “cognitive behav*” OR “structural modification*” OR “low-risk guideline*” OR “self-regulation” OR “notification*”)

Supplementary search methods

A Google Scholar search in Round 3 (February–May 2024) served as a sensitivity check for recently indexed papers. It was not a primary retrieval source. The few records it contributed were imported to Rayyan and screened against the same eligibility criteria as the database yield. Grey literature and conference abstracts remained excluded per the review protocol.

Citation chasing was conducted on the reference lists of all included papers and key reviews. Records were imported to Rayyan and screened against the same criteria.

A priori reference set used for search validation

The following papers were used to validate string sensitivity across all three rounds:

- Auer, M., & Griffiths, M. D. (2013). Voluntary limit setting and player choice in most intense online gamblers: An empirical study of gambling behaviour. *Journal of Gambling Studies*, 29(4), 647–660.
- Blaszczynski, A., Ladouceur, R., & Shaffer, H. J. (2004). A science-based framework for responsible gambling: The Reno model. *Journal of Gambling Studies*, 20(3), 301–317.
- Gainsbury, S. M. (2014). Review of self-exclusion from gambling venues as an intervention for problem gambling. *Journal of Gambling Studies*, 30(2), 229–251.

- Harris, A., & Griffiths, M. D. (2017). A critical review of the harm-minimisation tools available for electronic gambling. *Journal of Gambling Studies*, 33(1), 187–221.
- Ivanova, E., Rafi, J., Lindner, P., & Carlbring, P. (2019). Experiences of responsible gambling tools among non-problem gamblers. *Addictive Behaviors Reports*, 9, 100161.
- Ladouceur, R., Blaszczynski, A., & Lalande, D. R. (2012). Pre-commitment in gambling: A review of the empirical evidence. *International Gambling Studies*, 12(2), 215–230.
- Note.* Full eligibility criteria are documented in 01.2. Detailed PRISMA-ScR flow diagram with paper-level exclusion reasons is provided in 01.4. The screening record with record-level data is available in 01.5. RG = responsible gambling; PRISMA-ScR = Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for Scoping Reviews.